

# IRTc: Copula-Based Item Response Theory

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# **Chapter 1**

## **IRTc: Copula-Based Item Response Theory**

# Chapter 2

## About this thesis

This is the landing page for Damian Betebenner’s dissertation, “IRTc: Copula-Based Item Response Theory”, at dataimago (Independent research — educational measurement & psychometrics (dataimago)).

The thesis is rendered via Quarto + XeLaTeX to a PDF whenever you push changes to `ui/www/` or `R/`; a stable copy is committed to `docs/thesis.pdf` by the `build-thesis.yml` workflow.

### 2.1 Chapters

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## 2.2 Formatting

See [THESIS-CLS-README.md](#) for the 3-mode thesis class file strategy + how to iterate on your `thesis.cls` for institutional compliance.

# **Chapter 3**

## **Front matter**



# Abstract

Your dissertation abstract — typically one paragraph, ~250–350 words depending on institutional convention. Pull from your `dataimago-spec.yaml`'s `vertical.dissertation.thesis.contribution` field as a starting point.

# Acknowledgments

Acknowledgments for your advisor, committee, family, funders. If you have funding to acknowledge, also fill in your `dataimago-spec.yaml`'s funding section — that field can be used to auto-render this section in a future iteration.

# Dedication

Optional. Remove this section if not used.

# Chapter 4

## Introduction

### 4.1 Background

Conventional IRT treats local independence as the default assumption: conditional on ability, item responses factorize into a product of item-level probabilities. The founding conjecture of this project — treated strictly as a hypothesis, not evidence (claim C0001 in the ledger) — is that if Lord-era IRT had had access to Sklar’s theorem, copula theory, and modern compute, parts of IRT might have developed as a dependence-modeling framework in which local independence is simply the independence-copula case rather than the default.

*Expand from `dataimago-spec.yaml` □ `vertical.dissertation.thesis.background`: the intellectual genealogy (student growth percentiles, large-scale assessment, local dependence and testlets as persistent practical realities) and the dual-purpose framing (the dissertation as the first end-to-end stress test of the dataimago research machine).*

### 4.2 Research question

Can a copula-first formulation of Item Response Theory (IRTc) recover conventional local-independence IRT as the independence-copula special case — and on which real item-response datasets does separating item-response marginals from dependence structures improve held-out prediction, model fit, and diagnostic interpretability relative to Rasch/2PL/3PL/GRM/MIRT baselines?

## 4.3 Contribution

1. **Theory** — a formally specified IRTc model family: marginals separated from dependence structures, identifiability assumptions for discrete responses stated explicitly, and a precise characterization of which conventional IRT models are special cases, approximations, or mere analogies.
2. **Empirics** — fair, reproducible benchmarks against conventional baselines on simulated and Item Response Warehouse (IRW) data, with failure cases documented as first-class results.
3. **Infrastructure** — a reproducible research application in which every material claim is citation-backed via a claim ledger and every experiment is registered before it runs.

## 4.4 Roadmap

Chapter 2 frames the historical counterfactual. Chapter 3 examines conventional IRT and the local independence assumption. Chapter 4 introduces copulas and Sklar's theorem. Chapter 5 confronts the discrete response problem. Chapter 6 defines the IRTc model family and locates conventional IRT within it. Chapter 7 develops estimation strategies and simulation studies. Chapter 8 presents IRW benchmark datasets and results. Chapter 9 closes with diagnostics, sensitivity analyses, and limitations.

# Chapter 5

## Historical Framing: From Empirical Curves and Normal Ogives to Dependence Modeling

### 5.1 The empirical-curve and normal-ogive lineage

*Trace the development from empirical item characteristic curves through normal-ogive and logistic models (Lord, Novick, Rasch, Birnbaum, Samejima, Bock) — primary sources per the WP1 bibliography; every historical claim carries a source ID.*

### 5.2 Local independence as a founding assumption

*How and why conditional independence became the default: computational constraints, estimation tractability, and the era's available mathematics.*

### 5.3 The counterfactual, stated carefully

*The Lord-era conjecture (claim C0001, hypothesis): a generator of research questions, never evidence. What the counterfactual does and does not license — per the claim-ledger rules, this chapter frames questions; it proves nothing.*

## 5.4 Dependence modeling arrives

*Sklar (1959) to modern copula families (Joe, Nelsen); factor-copula item response models (Kadhem & Nikoloulopoulos 2021) as the closest existing work.*

# Chapter 6

## Conventional IRT and the Local Independence Assumption

### 6.1 The conventional model family

*Rasch/1PL, 2PL, 3PL, graded response, (generalized) partial credit, MIRT and bi-factor — the Level-0 baselines of the model ladder, implemented via `mirt` and `TAM`.*

### 6.2 Local independence, formally

For a response vector  $Y_i = (Y_{i1}, \dots, Y_{iJ})$ :

$$P(Y_i = y_i \mid \theta_i) = \prod_j P(Y_{ij} = y_{ij} \mid \theta_i, \xi_j)$$

*State the assumption precisely; distinguish it from unidimensionality.*

### 6.3 Where the assumption strains

*Local dependence, testlets, item bundles, speededness, multidimensionality — the persistent practical realities conventional models treat as nuisances. Q3-style residual diagnos-*



*tics and limited-information fit statistics as the detection toolkit.*

## **6.4 Existing accommodations**

*Testlet response theory, bi-factor/two-tier models, latent class IRT — what each concedes to dependence, and at what interpretive cost.*

# Chapter 7

## Copulas, Sklar's Theorem, and Dependence Structures

### 7.1 Copulas and Sklar's theorem

*Definitions, Sklar's theorem, and the uniqueness caveat: uniqueness holds under continuous marginals — a caveat that becomes central in Chapter 5.*

### 7.2 Copula families and their dependence signatures

*Gaussian/BVN, Student t, Frank, Clayton/Gumbel and rotations, Joe/BB families: tail dependence, asymmetry, and what each signature could mean psychometrically (items jointly failing among low-ability examinees; region-specific discrimination).*

### 7.3 Structured constructions

*Factor copulas, bi-factor and second-order copulas, vines, factor trees — the constructions the model ladder's Levels 2–4 build on (FactorCopula as the implementation reference; capability claims only).*

# Chapter 8

## The Discrete Response Problem

*The load-bearing chapter for mathematical honesty — see [wiki/theory/discrete-copula-cautions.m](#) written before any model claims.*

### 8.1 Why discreteness breaks naive Sklar arguments

*Item responses are dichotomous or ordinal; the copula linking observed responses is not unique (claim C0002). No “mathematically cleaner” claim may lean on continuous-Sklar uniqueness.*

### 8.2 Candidate resolutions

1. **Latent-continuous formulation** — categories as thresholds of latent continuous response variables; copulas on the latent scale.
2. **Exact rectangle-probability likelihoods** — category probabilities as copula rectangle differences.
3. **Bayesian data augmentation** — Smith & Khaled (2012) as the anchor caution: estimation difficulty beyond the bivariate case.
4. **Parametric factor-copula constructions** — existing discrete-data factor copula models as the starting point.
5. **Distributional/randomized transforms** — only if the literature supports them for the target inference.

### 8.3 Identifiability conditions

*State what is and is not identified for dichotomous and ordinal data under each resolution — every theorem-like statement gets proof, citation, or a conjecture label (WP2 gates).*

# Chapter 9

## The IRTc Model Family and Conventional IRT as Special Cases

### 9.1 The IRTc construction

$$P(Y_i \leq y_i \mid \theta_i) = C_\theta(F_1(y_{i1} \mid \theta_i, \xi_1), \dots, F_J(y_{iJ} \mid \theta_i, \xi_J); \delta)$$

*Item-response marginals  $F_j$  separated from the dependence structure  $C_\theta$  with parameters  $\delta$ ; copula placement options (among observed responses conditional on  $\theta$ , among latent response variables, among latent dimensions, among residuals, among testlet effects).*

### 9.2 Conventional IRT located within IRTc

*The independence copula recovers local-independence IRT (Level 1 of the model ladder — the product-copula wrapper with its numerical-tolerance acceptance test). For each conventional model, classify the relationship: exact equivalence, approximation, or analogy — nothing stronger than what is proved.*

### 9.3 Parameter interpretation under dependence

*What difficulty, discrimination, guessing, and thresholds mean when dependence is no longer Gaussian or product-form; when a copula parameter is a substantive estimand vs a nuisance parameter for fit; whether any Rasch-style invariance survives restricted copula families.*

# Chapter 10

## Estimation Strategies and Simulation Studies

### 10.1 Estimation strategies

*Per model-ladder level: `mirt`/TAM baselines (EM, MHRM); FactorCopula estimation for one-factor and structured copulas; rectangle likelihoods; Bayesian augmentation where justified. Every fit records versions, seeds, optimizer settings, convergence diagnostics, and warnings (WP3 gates — no silent inclusion of nonconverged fits).*

### 10.2 Simulation designs

*Every model has a simulation recipe before touching real data (WP2 gate): binary 2PL and ordinal GRM generators first, then copula-family generators for parameter recovery.*

### 10.3 Parameter recovery

*Recovery RMSE and bias by copula family, sample size, and item count; stability across optimization starts.*

## 10.4 The Level-1 equivalence test

*The product-copula wrapper must match baseline estimates, log-likelihoods, and predicted probabilities within numerical tolerance — differences explained and documented, or the ladder does not advance.*



# Chapter 11

## IRW Benchmark Datasets and Results

### 11.1 The Item Response Warehouse

*IRW v28.2: 900+ harmonized open item-response datasets, 2B+ responses (Behavior Research Methods, 2025). Access via the `irw` R package with Redivis authentication. Not every IRW dataset suits every IRTc model — selection criteria and data cards per WP4.*

### 11.2 Benchmark design

*Tiered datasets (smoke-test / primary / stress-test); immutable raw zone; reproducible splits; fairness rules from charter §14.3 — same preprocessing, same splits, same inclusion criteria, failed fits reported, stronger baselines before any IRTc gain is claimed.*

### 11.3 Results

*Held-out log loss and Brier score as primary metrics; marginal likelihood, AIC/BIC/SABIC, M2, calibration by ability strata as secondary. Every table carries model version, dataset version, split ID, and seed. Results appear here only from pre-registered experiments (`experiments/registry.yaml`).*

## 11.4 Failure cases

*Documented as first-class results — datasets where IRTc does not improve on baselines are reported with the same prominence as successes.*

# Chapter 12

## Diagnostics, Sensitivity Analyses, and Limitations

### 12.1 Diagnostics

*Residual-dependence (Q3-style) reduction, tail-dependence diagnostics, local-dependence heatmaps, calibration curves, ability-estimate comparisons — can IRTc diagnostics see what conventional residual diagnostics miss?*

### 12.2 Sensitivity analyses

*Copula family set, starting values, quadrature/integration method, sample and item sub-sampling, missingness handling, category collapsing, group/testlet definitions, split construction, optimizer thresholds (chapter §14.4). Sensitivity precedes any superiority claim (WP5 gate).*

### 12.3 Interpretability and stability

*Are the additional dependence parameters stable and interpretable at realistic educational-testing sample sizes, or does model-selection risk erase the gains?*

## 12.4 Limitations

*Required by the claim-ledger rules: where IRTc is not yet justified, where identifiability remains conjectural, where compute constrains the ladder, and what the acceptance criteria (charter §20) still leave open.*

## **Chapter 13**

### **Back matter**

# References

The bibliography is rendered automatically from `../references.bib`. No content needed here — Quarto handles it.

# Appendices

Optional. If you have appendices (data tables, supplementary analyses, code listings, interview protocols), add them as `chapters/99a-appendix-X.qmd`, `99b-appendix-Y.qmd`, etc. and reference them in `../_quarto.yml`.